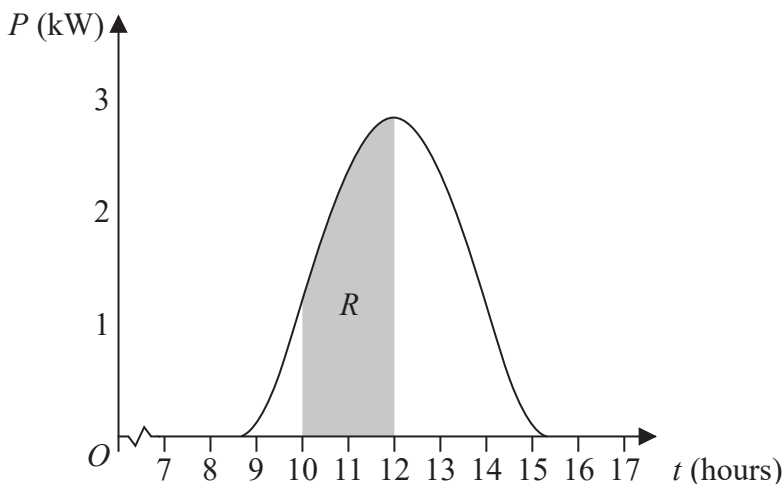


7.



**Figure 1**

Solar panels are installed on the roof of a building.

The power,  $P$ , produced on a particular day, in kW, can be modelled by the equation

$$P = 0.95 + 2^{t-12} + 2^{12-t} - (t-12)^2 \quad 8.5 \leq t \leq 15.2$$

where  $t$  is the time in hours after midnight. The graph of  $P$  against  $t$  is shown in Figure 1.

A table of values of  $t$  and  $P$  is shown below, with the values of  $P$  given to 4 significant figures where appropriate.

|                      |    |       |      |      |      |
|----------------------|----|-------|------|------|------|
| Time, $t$<br>(hours) | 10 | 10.5  | 11   | 11.5 | 12   |
| Power, $P$<br>(kW)   |    | 1.882 | 2.45 |      | 2.95 |

- (a) Use the given equation to complete the table, giving the values of  $P$  to 4 significant figures where appropriate.

(2)

The amount of energy, in kWh, produced between 10:00 and 12:00 can be found by calculating the area of region  $R$ , shown shaded in Figure 1.

- (b) Use the trapezium rule, with all the values of  $P$  in the completed table, to find an estimate for the amount of energy produced between 10:00 and 12:00. Give your answer to 2 decimal places.

(4)